

RAG-Powered Knowledge

Agents 1: Embeddings, Chunking & Indexing

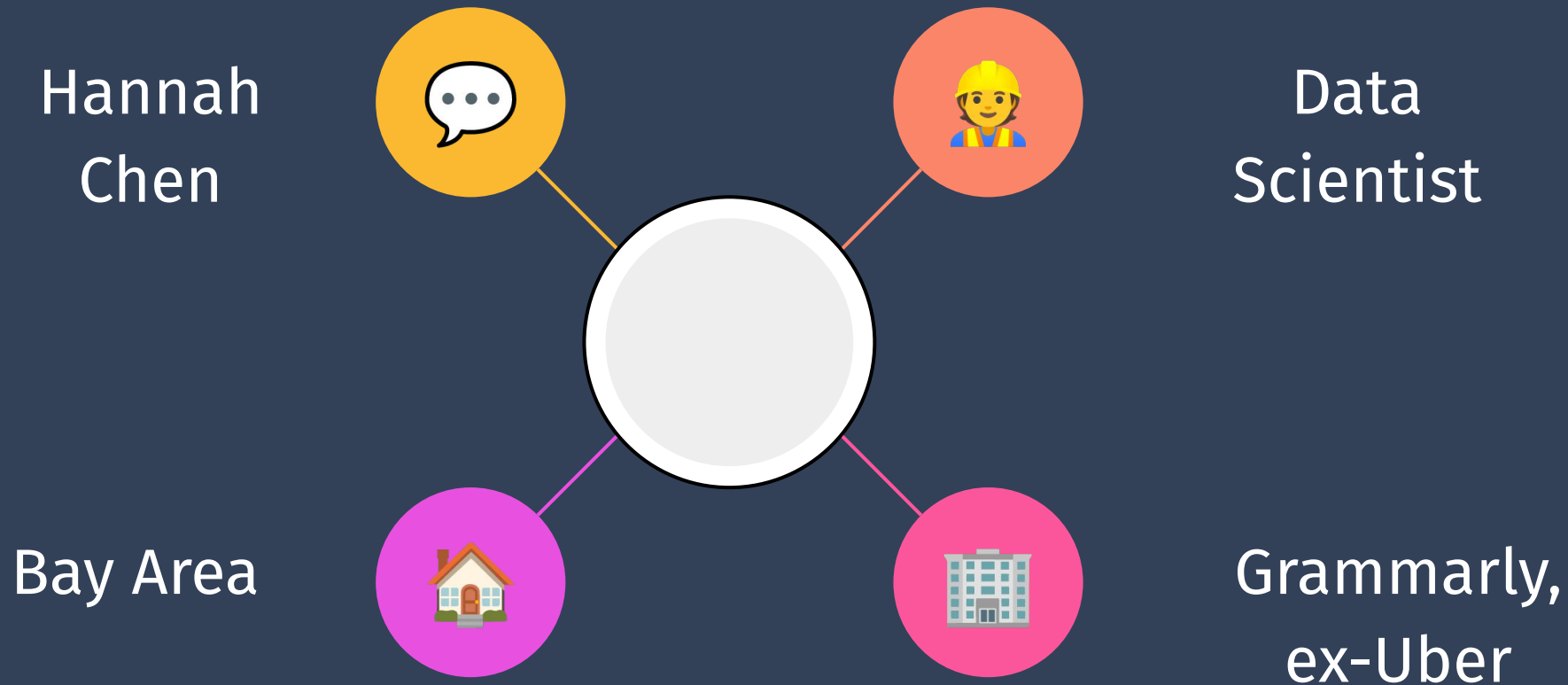


Introduction - Mike Conti

- Senior Software Engineer
- ex-Amazon AGI (Artificial General Intelligence)
- BA Computer Science – University of Maryland, College Park
- <https://www.linkedin.com/in/mike-conti/>



Welcome to today's class, before we begin,
pop into the chat:



Structure of Class

1. **Sunday Class - 4 hr (Sunday)**
2. **Assignments** - MCQ & Coding Assignment



Optimize Your Experience

- ✓ Interact with your instructors via Sunday live class.
- ✓ Don't be shy to speak up and get clarifications !



- ✓ Solve your assignments, MCQs and other assessments and get feedback (Thursday review session)
- ✓ Use your resources! It's your experience – what you put in is what you'll get out.

Don't worry!

We are here to help at your every step !

But, how???



Are you using all these IK support features?

- **Uplevel** - contains all the learning resources - videos, MCQs, assignments
- Optional **Post class videos** for additional reference
- **Technical coaching sessions** on Wednesday to clarify all your queries
- **Discord** community for peer learning
- **TAs** to help answer the queries in the live class
- **Support tickets** - Raise them any time you want any clarifications/ tech concerns , etc on UpLevel

Success Hacks

- Be prepared for the class by watching the pre class videos
- Attend every class
- Participate in class
- Attempt all the assignments,
- Actively advocate for yourself
- Keep your goal in mind
- Consistency is the key

Be patient with yourself !

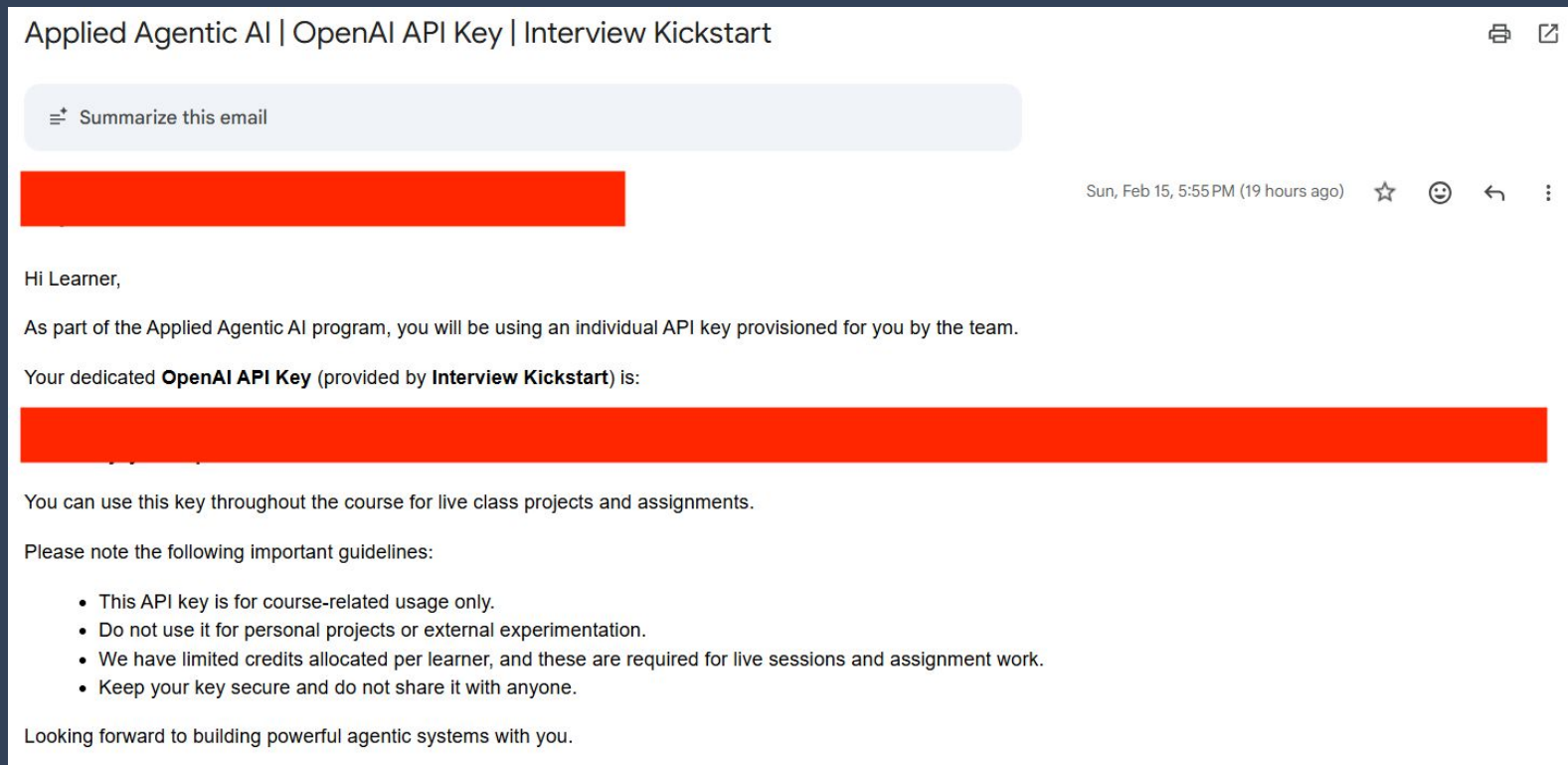
Check Your Email for API Key

You should have received an email like the one shown below containing your OpenAI API key.

If you do not see this email:

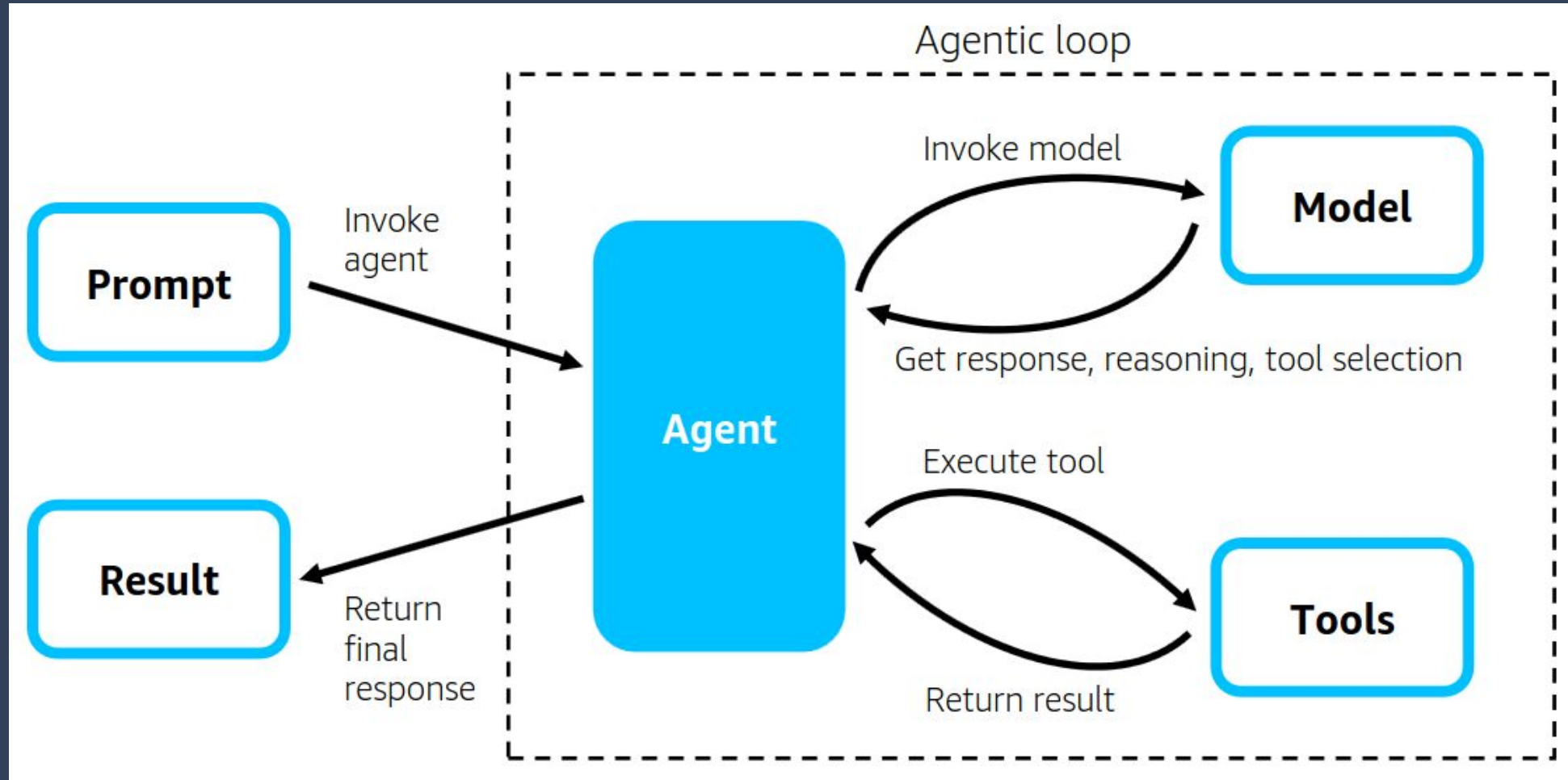
- Please check your inbox
- Also check your spam or junk folder

If you still cannot find it, please reach out to the Operations Team operations@interviewkickstart.com



Week 1 Recap: Agentic AI Foundations

Recap: The Anatomy of an Agentic loop



Today's Agenda



1



Text
Embeddings

2



Chunking
Strategies

3



Indexing
Strategies
for RAG

**Let's understand these concepts
through a hands-on project**

Clone & Practice: Your Learning Repository



SupportDesk-RAG-Workshop

<https://github.com/jeev1992/SupportDesk-RAG-Workshop>

What's Included:



6 Complete Modules

Embeddings, Chunking, Indexing, RAG Pipeline, Evaluation, Agentic RAG



Working Code Examples

Ready-to-run demos, practice exercises, and solutions for each module



Getting Started

```
git clone https://github.com/jeev1992/SupportDesk-RAG-Workshop.git
```

1. Install dependencies:
`pip install -r requirements.txt`
2. Configure OpenAI API key
3. Run demos from each module directory

Note: The following modules will be covered in **Week 3: RAG Pipeline, Evaluation, and Agentic RAG.**

Problem statement: What exactly are we building?

In Week 2, we are building the retrieval layer for the **SupportDesk RAG** assistant, the part of the system that can reliably find the most relevant past support tickets for any new issue.

The problem we're solving (in the real world)

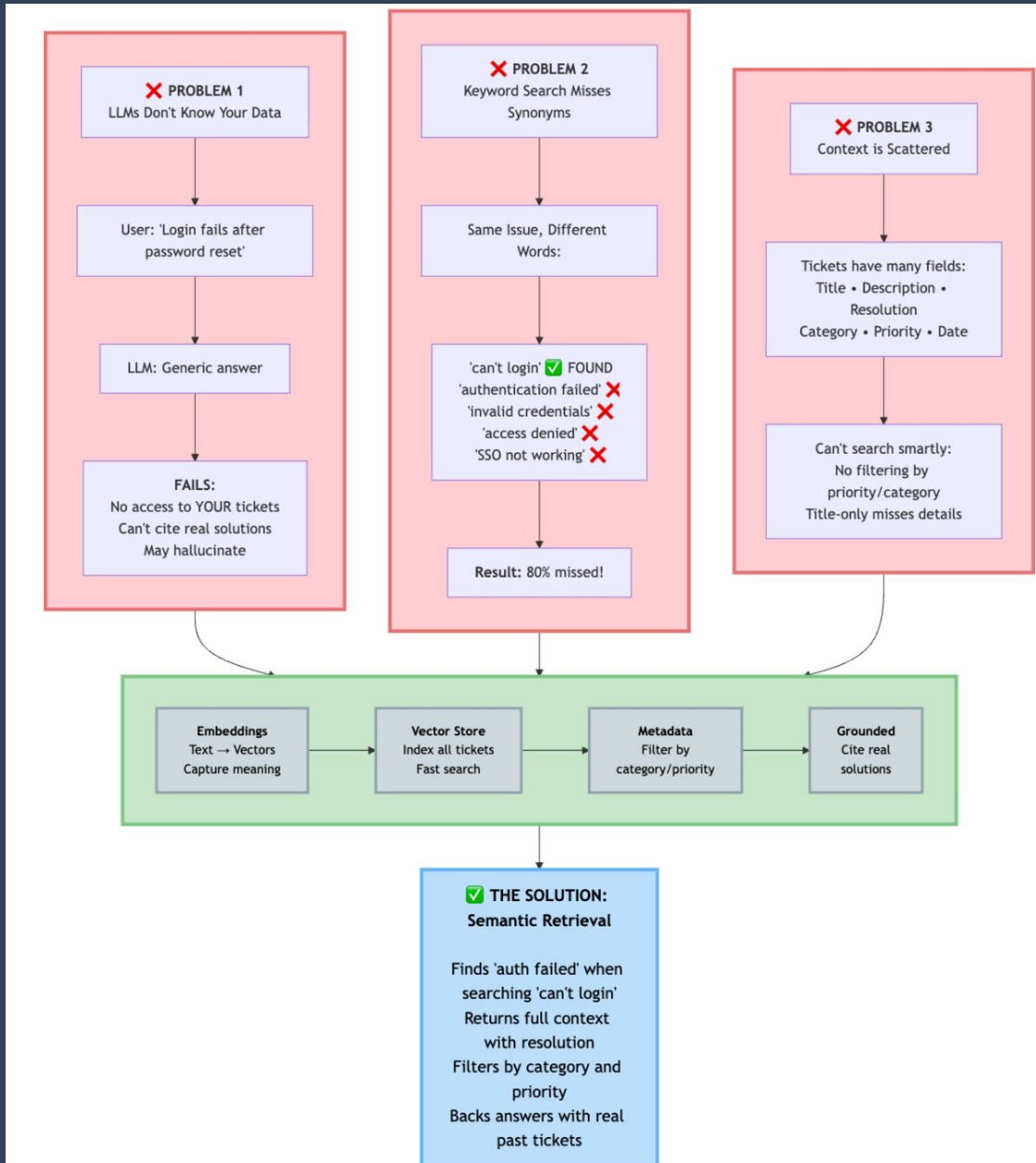
Engineers and Support teams and regularly face issues like:

- Users can't log in after password reset
- Database connection timeouts in production
- Payments failing for international cards
- Emails not being delivered

The challenge is: **the solution often already exists in past tickets**, but finding it is slow because:

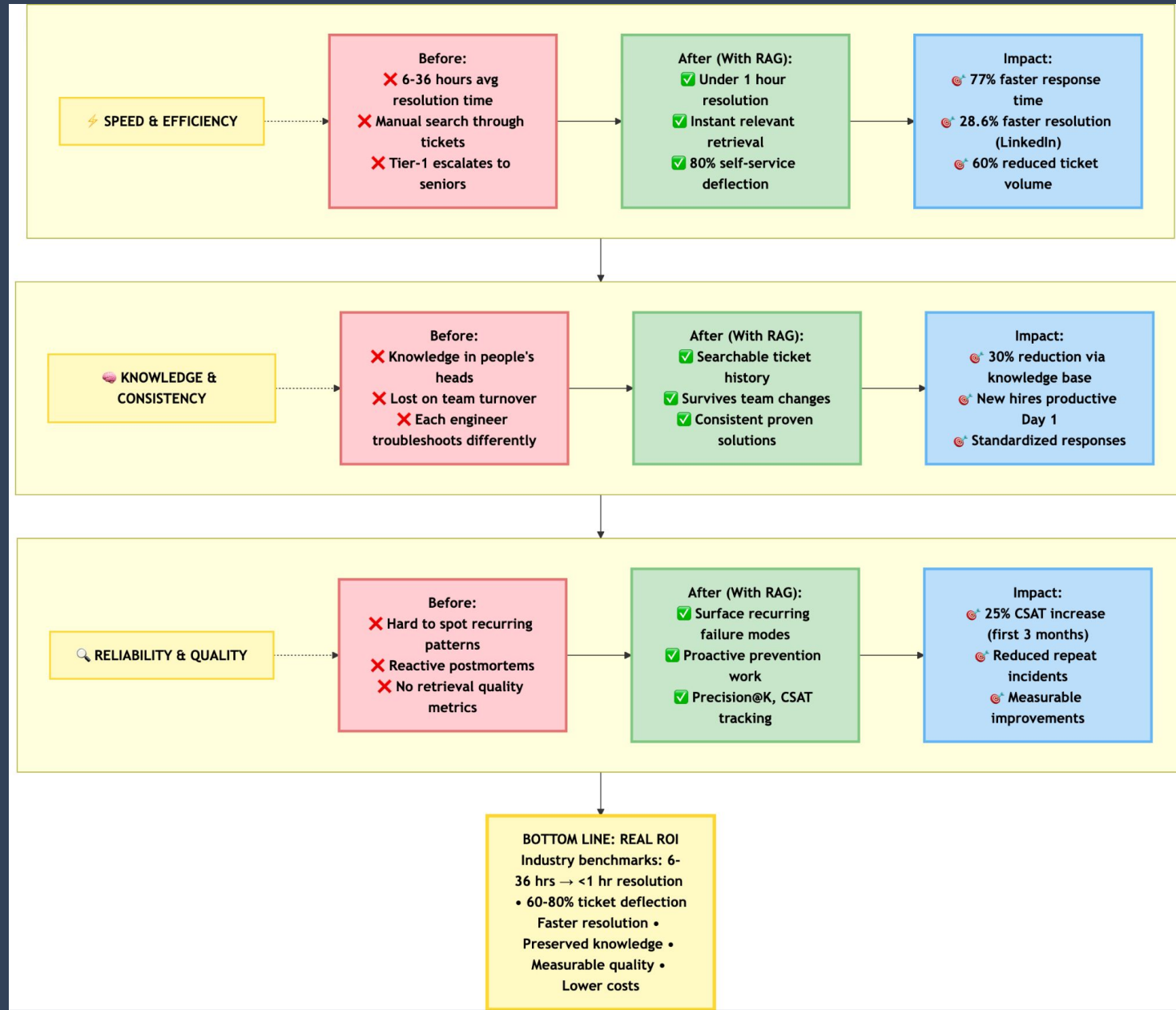
- Ticket data is **unstructured text** (not clean tables)
- People describe the same issue in **different words**
- Search is often **keyword-based**, so it misses relevant results
- There are too many tickets to read manually

Why We Need Semantic Retrieval: The 3 Critical Gaps



- **LLM-only approach:** can sound correct but **may guess/hallucinate**, and it doesn't know your ticket history by default.
- **Context limits:** we can't paste all tickets into the prompt; we must retrieve only the best few.
- **Noise vs precision trade-off:** without chunking + indexing choices, retrieval becomes inconsistent:

Why this matters for Engineering Teams



Example Queries

What Users Will Ask	What the System Sees
Query 1 Users can't log in after resetting their password.	 Target Match: TICK-001 (<i>Authentication</i>) Desired Output: This is caused by stale session tokens. The fix is to clear active sessions and force re-authentication.
Query 2 I'm seeing 402 errors on international payments.	 Target Match: TICK-003 (<i>Payment Gateway</i>) Desired Output: Enable the international processing flag in the gateway configuration.
Query 3 Why is the dashboard slow?	 Target Match: TICK-010 (<i>Performance</i>) Desired Output: It is likely an N+1 query issue. Check if Redis caching is enabled for aggregate stats.

Let's Get It Started

what was the root cause resolution to ticket94 on the incident where someone spilled water on a computer in the warehouse



Show details ▾

The root cause resolution for **ticket 94** regarding the water spill in the warehouse was the installation of **clear acrylic "sneeze guards" or splash shields** over the terminal stations.

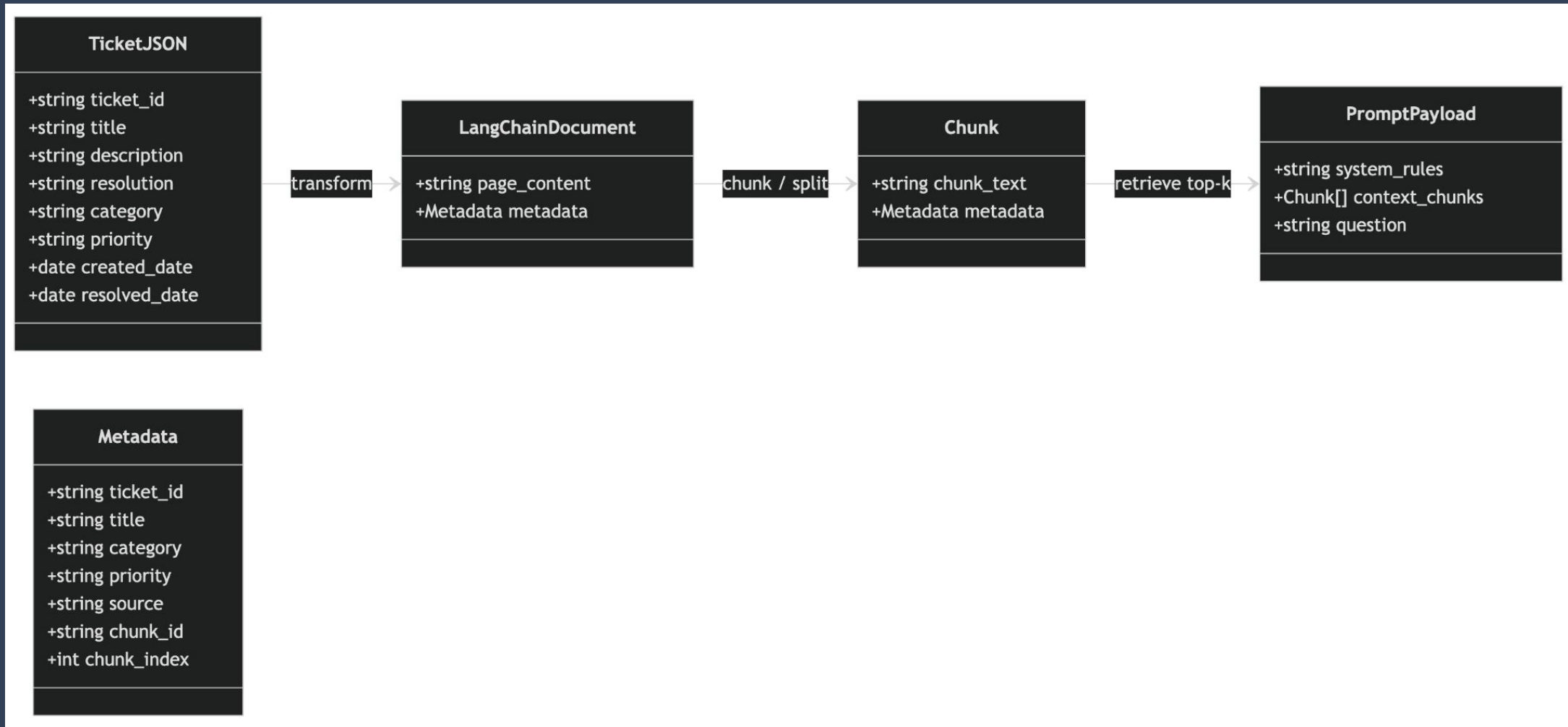
While the immediate fix was replacing the damaged hardware, the formal Root Cause Analysis (RCA) determined that the workstation's placement near a high-traffic hydration area made it vulnerable. To prevent a recurrence, the following steps were taken:

- **Engineering Control:** Mounted protective shields to deflect splashes.
- **Relocation:** The terminal was moved 3 feet away from the water dispenser.
- **Procedural Update:** Signage was added to designate the area as a "No Open Container" zone.

If you're looking for the specific technical logs or the name of the technician who closed it, those would be in the internal IT portal under the "Resolution" tab.

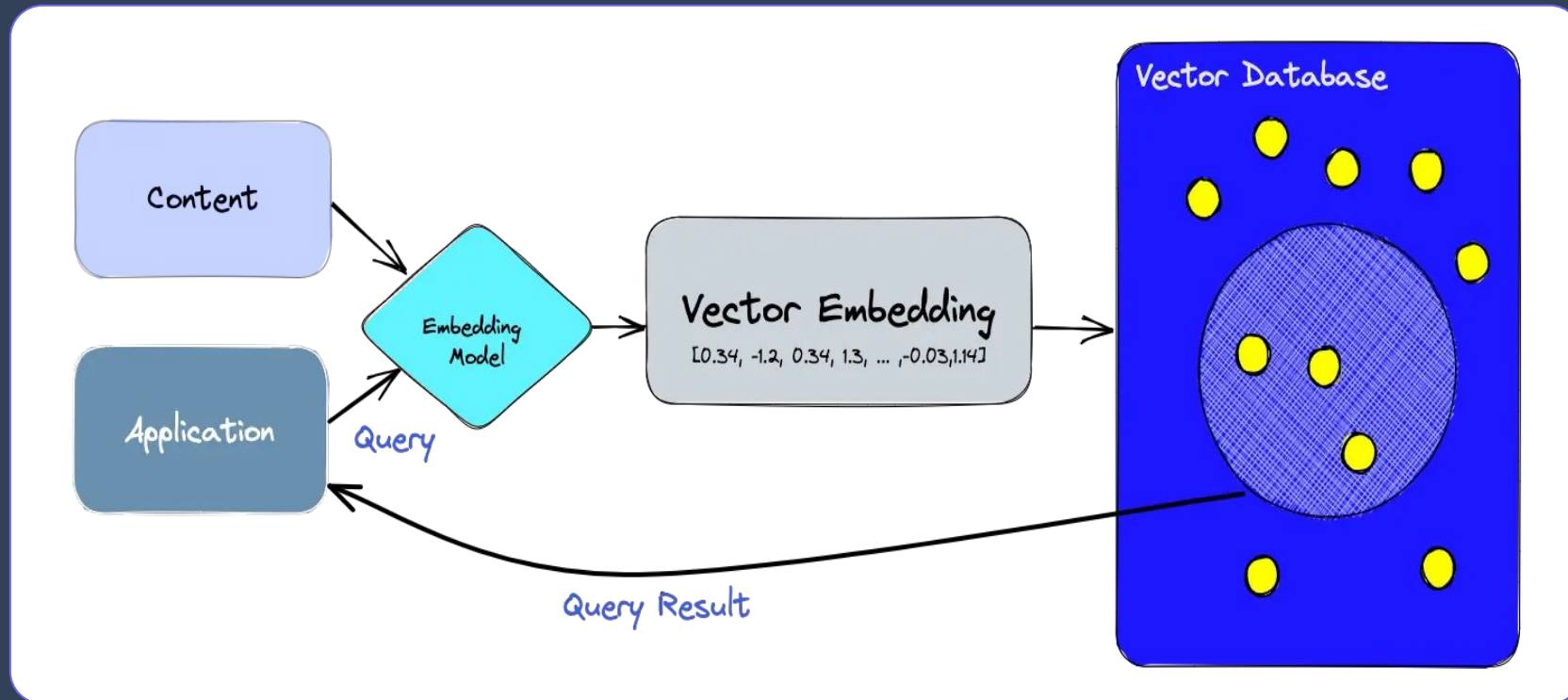
The Dataset

Data Schema (What the model sees):

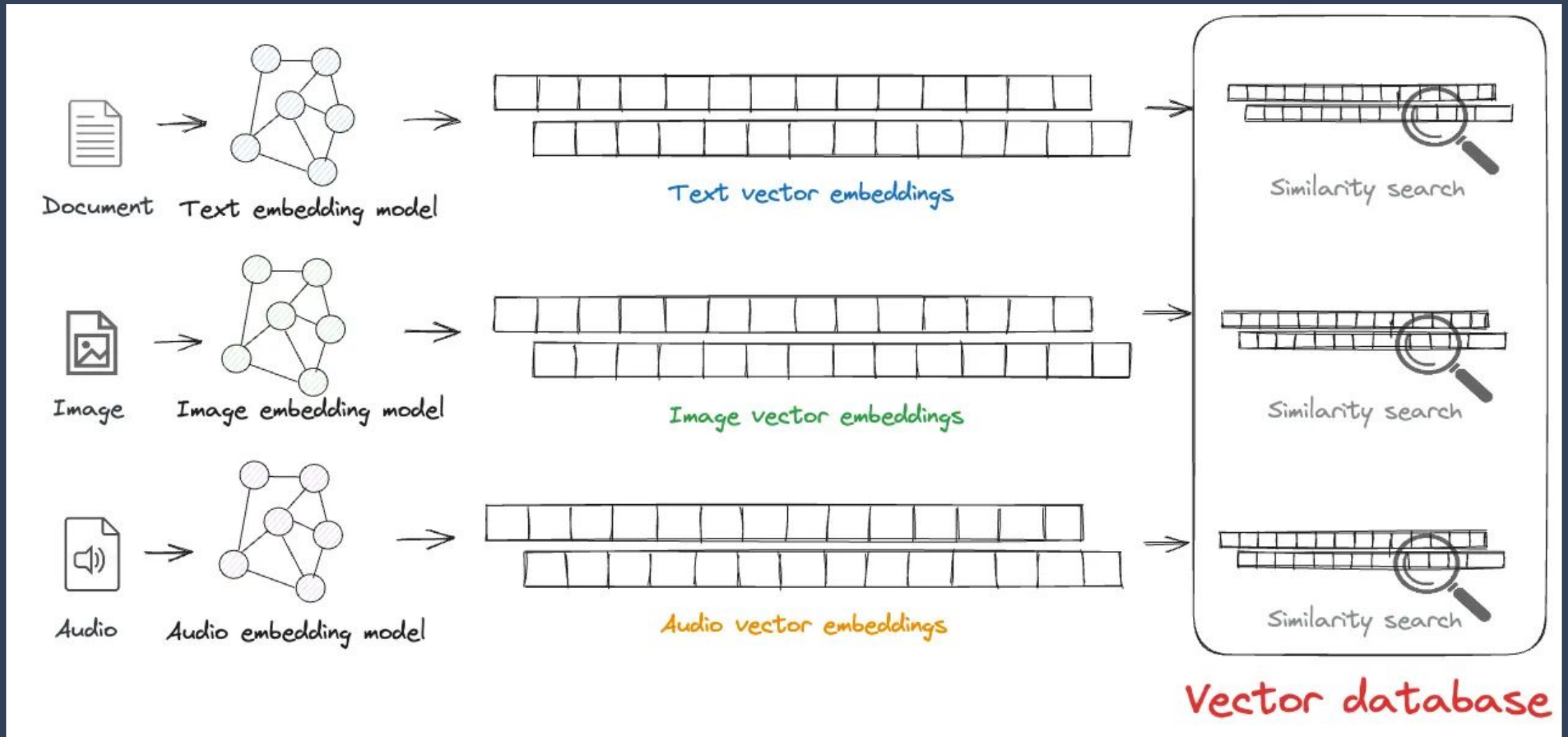


Text Embeddings: Understanding Text Embeddings

Vector Databases - Why you should know it?



Storing the embeddings in vector databases



Vector Store

- We know how to search for data in DB (SQL or NoSQL)
- But how to search for text data?



- **Can we do string match?**

Vector Store

- We know how to search for data in DB (SQL or NoSQL)
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- Can we do string match? - **Why not**

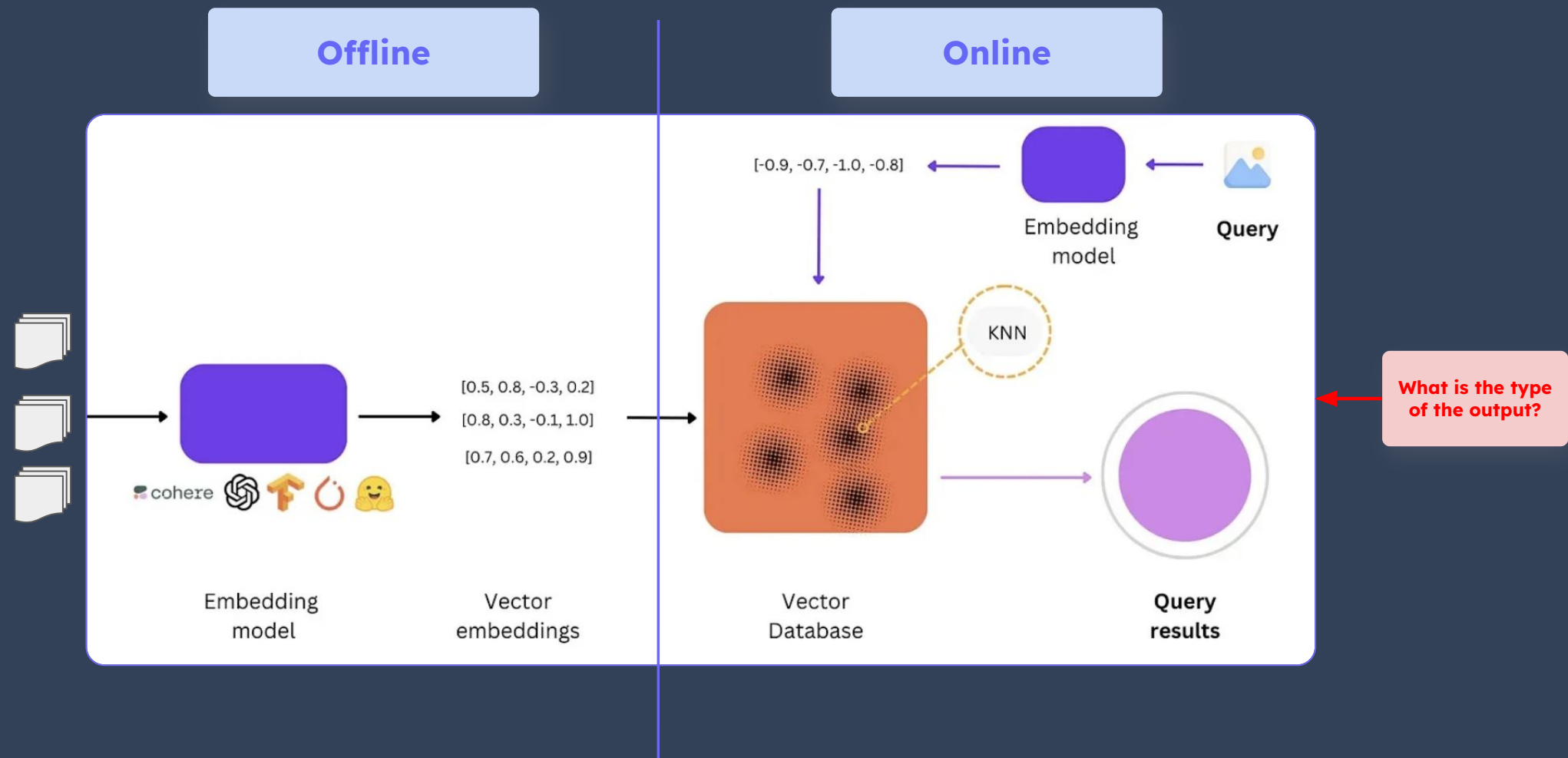
Vector Store

Can we do string match?

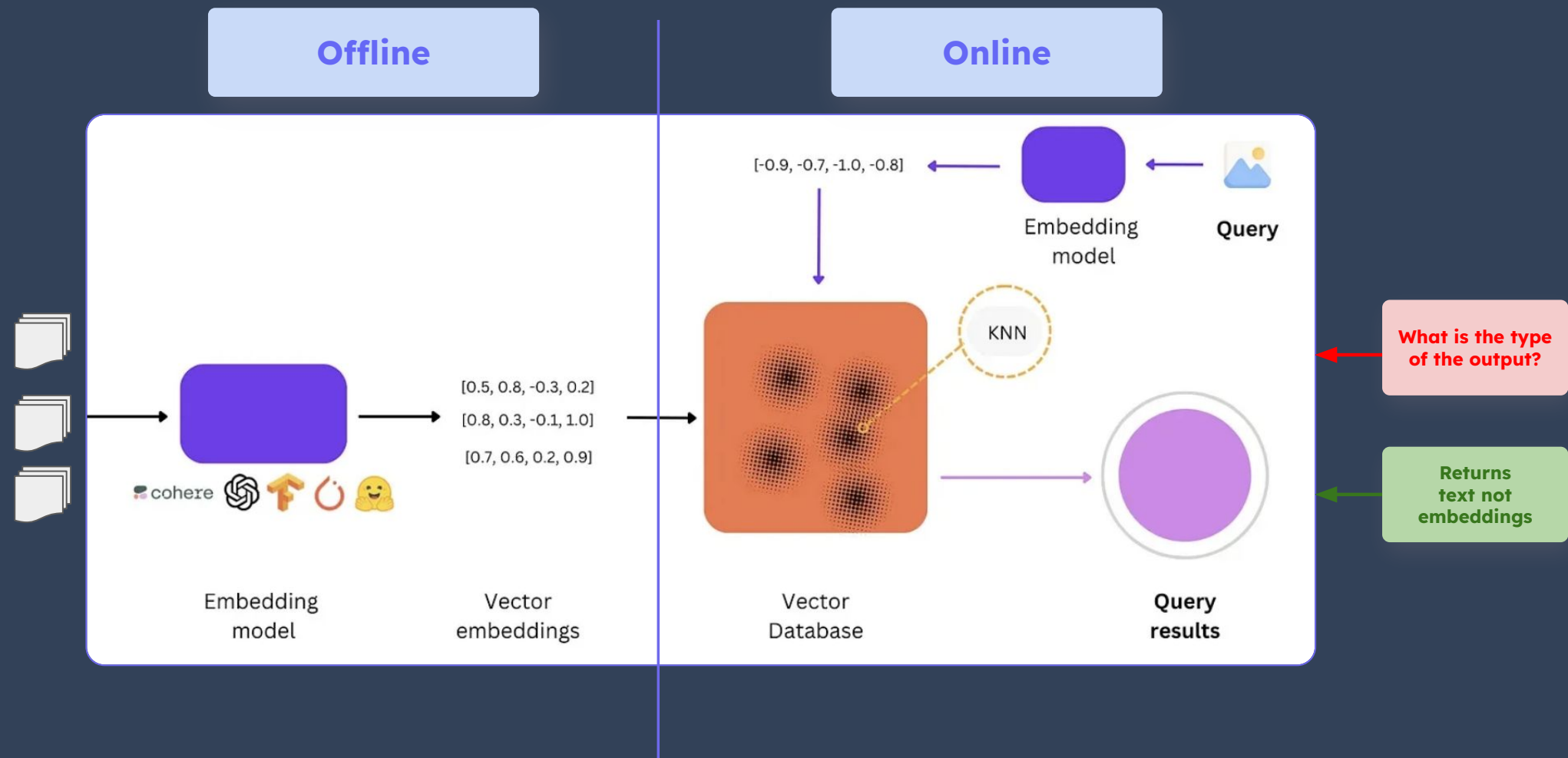
- Does not handle synonyms or rephrasing
 - “Car” vs “Vehicle”
- No semantic understanding
 - “I am travelling to Mexico, what phone plan should I use” vs “International plans”

What can we use instead?

Vector Store

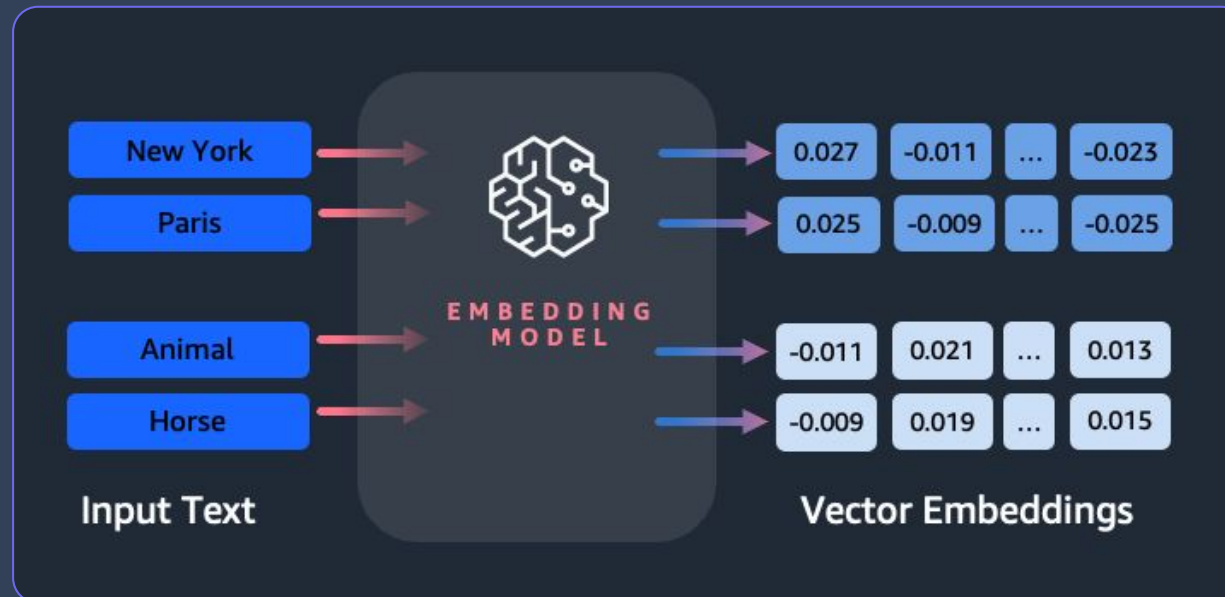


Vector Store

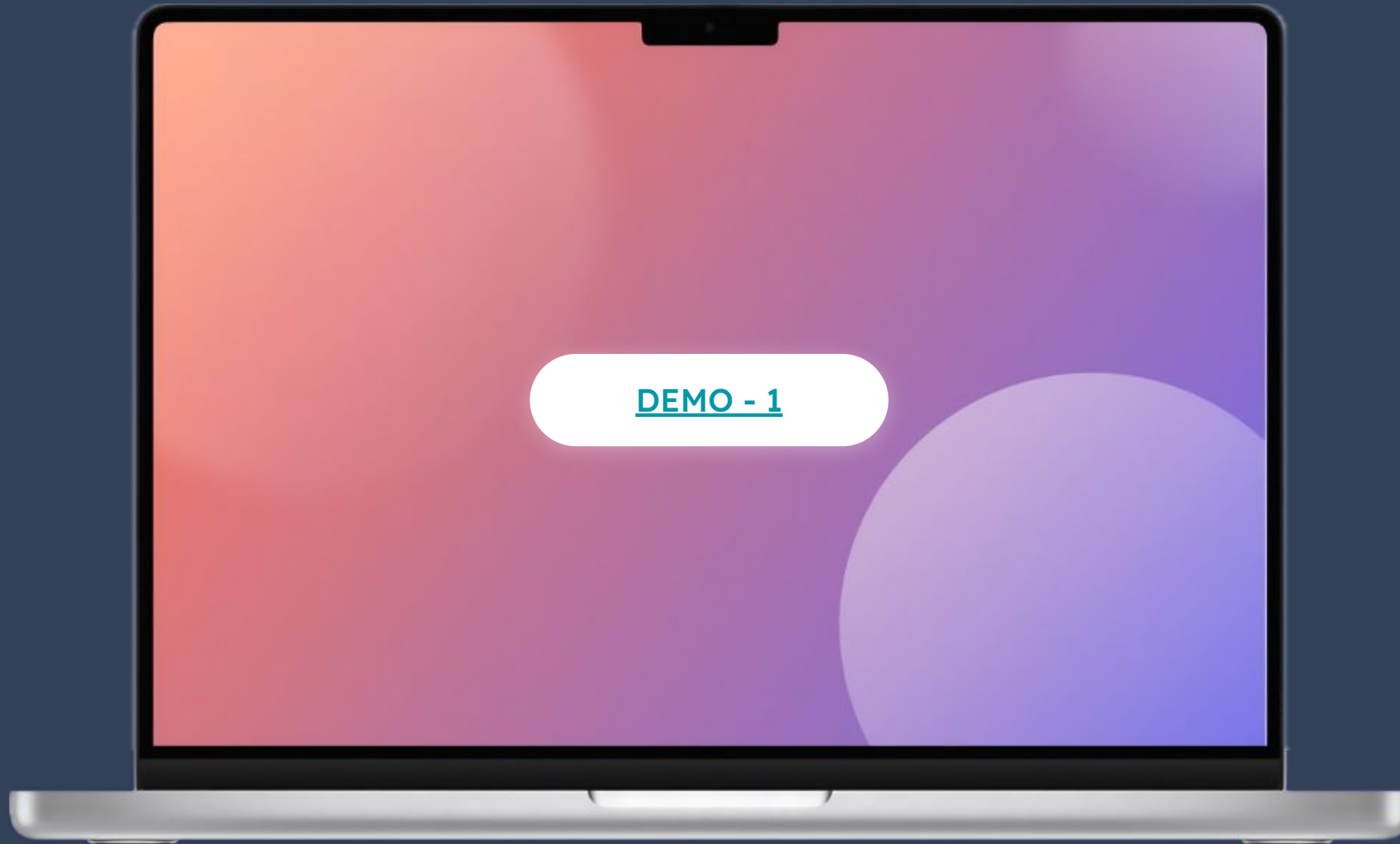


What is Embedding?

- Embeddings create a **vector representation** of a piece of text.
- We can think about text in the vector space
- Do things like semantic search
 - Where we look for pieces of text that are most similar in the vector space.



Module 1 - Implementation Notes: Text Embeddings



No Question Is Too Small Every Question Deserves an Answer!!

Ask about anything:

- Concepts from today's modules
- How this applies to your work
- Implementation details
- Clarifications on what we covered

Quick Request

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Chunking: Breaking Down Long Documents

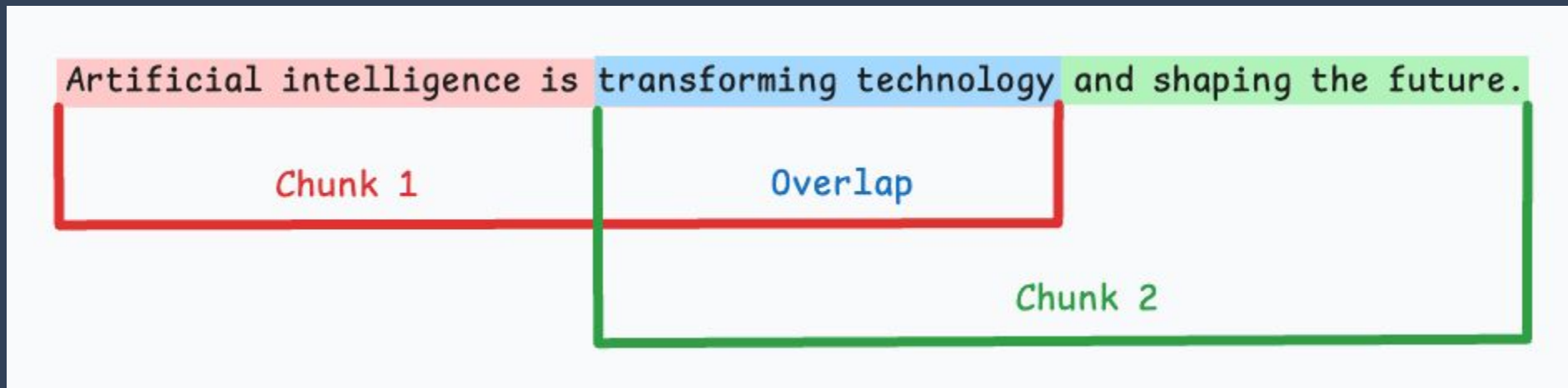


Chunking: Breaking Down Long Documents

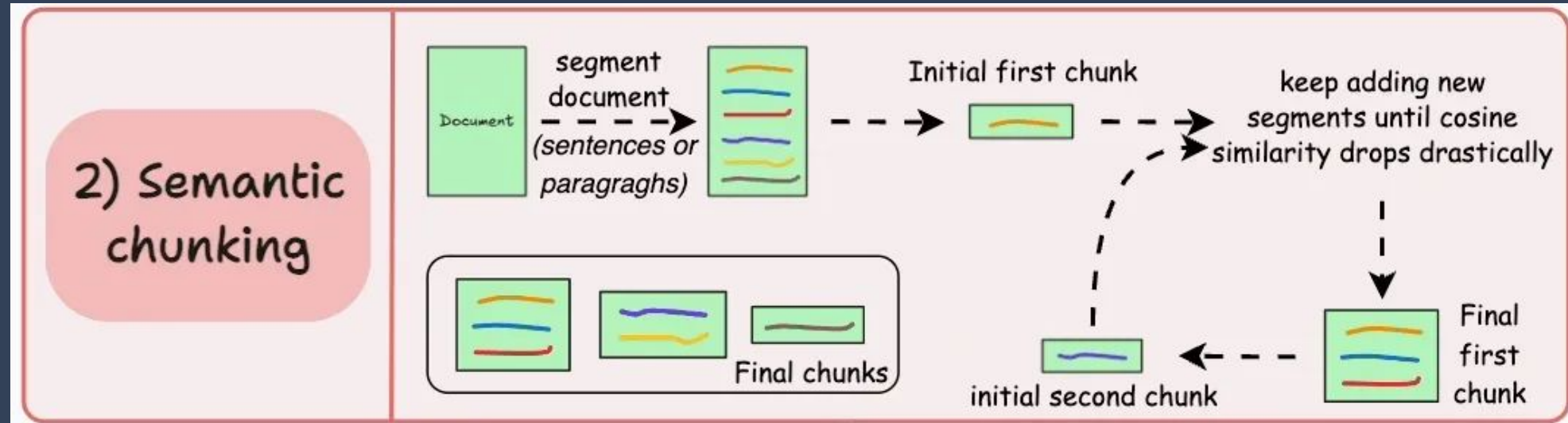


1) Fixed-size chunking

The most intuitive and straightforward way to generate chunks is by splitting the text into uniform segments based on a pre-defined number of characters, words, or tokens.



2) Semantic chunking

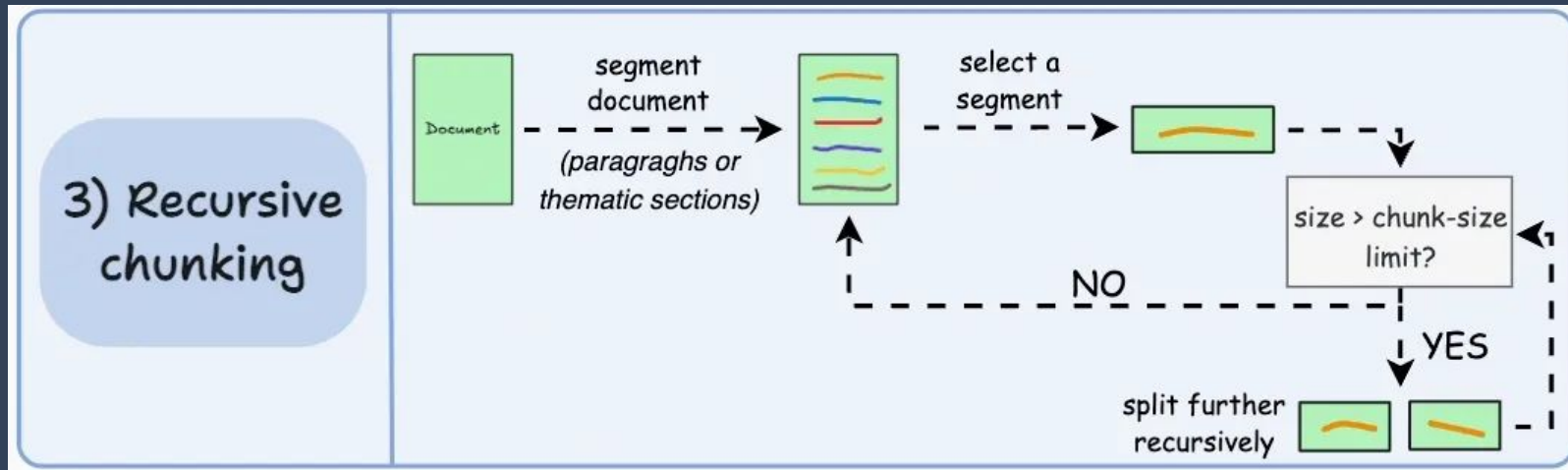


- Segment the document based on meaningful units like sentences, paragraphs, or thematic sections.
- Next, create embeddings for each segment.

Here's what the output could look like:

Artificial intelligence is transforming industries by automating processes, enhancing decision-making, and providing insights through data analysis. Machine learning, a subset of AI, enables systems to learn and improve from experience without explicit programming. Deep learning, a branch of machine learning, uses neural networks with multiple layers to model complex patterns in data.

3) Recursive chunking



First, chunk based on inherent separators like paragraphs, or sections. Next, split each chunk into smaller chunks if the size exceeds a pre-defined chunk size limit. If, however, the chunk fits the chunk-size limit, no further splitting is done.

Here's what the output could look like:

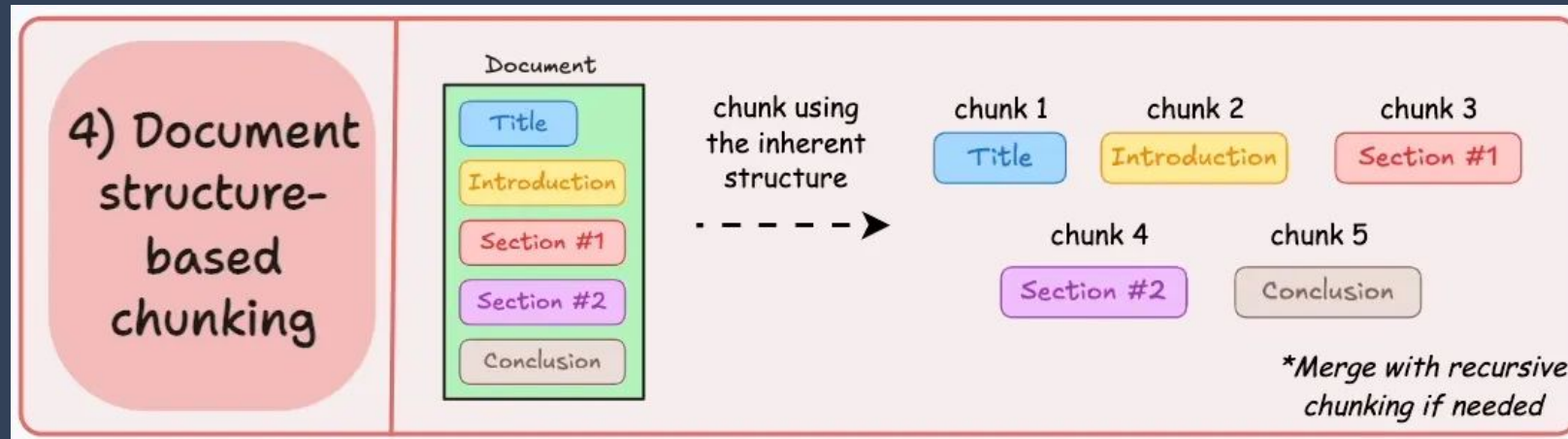
Paragraph 1

Artificial intelligence is transforming industries by automating processes, enhancing decision-making, and providing insights through data analysis. Machine learning, a subset of AI, enables systems to learn and improve from experience without explicit programming. Deep learning, a branch of machine learning, uses neural networks with multiple layers to model complex patterns in data.

Paragraph 2

AI is also improving natural language processing, enabling applications like chatbots and virtual assistants.

4) Document structure-based chunking

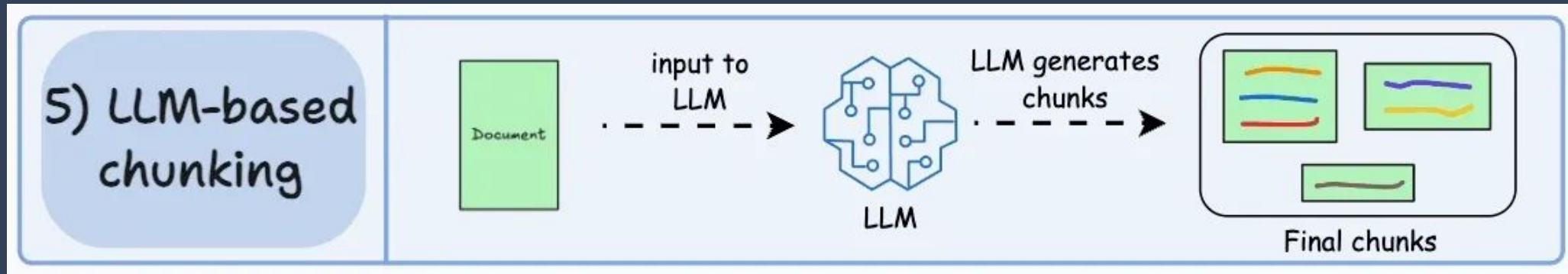


It utilizes the inherent structure of documents, like headings, sections, or paragraphs, to define chunk boundaries. This way, it maintains structural integrity by aligning with the document's logical sections.

Title: The Role of Artificial Intelligence in Modern Education	Chunk 1
Introduction	Chunk 2
Artificial intelligence (AI) is reshaping education by providing personalized learning experiences and automating administrative tasks.	
Section 1: Personalized Learning	Chunk 3
AI enables the customization of educational content to meet individual student needs, enhancing engagement and comprehension.	
Section 2: Administrative Automation	Chunk 4
From grading to scheduling, AI tools are streamlining administrative processes, allowing educators to focus more on teaching.	
Conclusion	Chunk 5
The integration of AI in education holds the promise of more efficient learning environments and improved student outcomes.	

Here's what the output could look like:

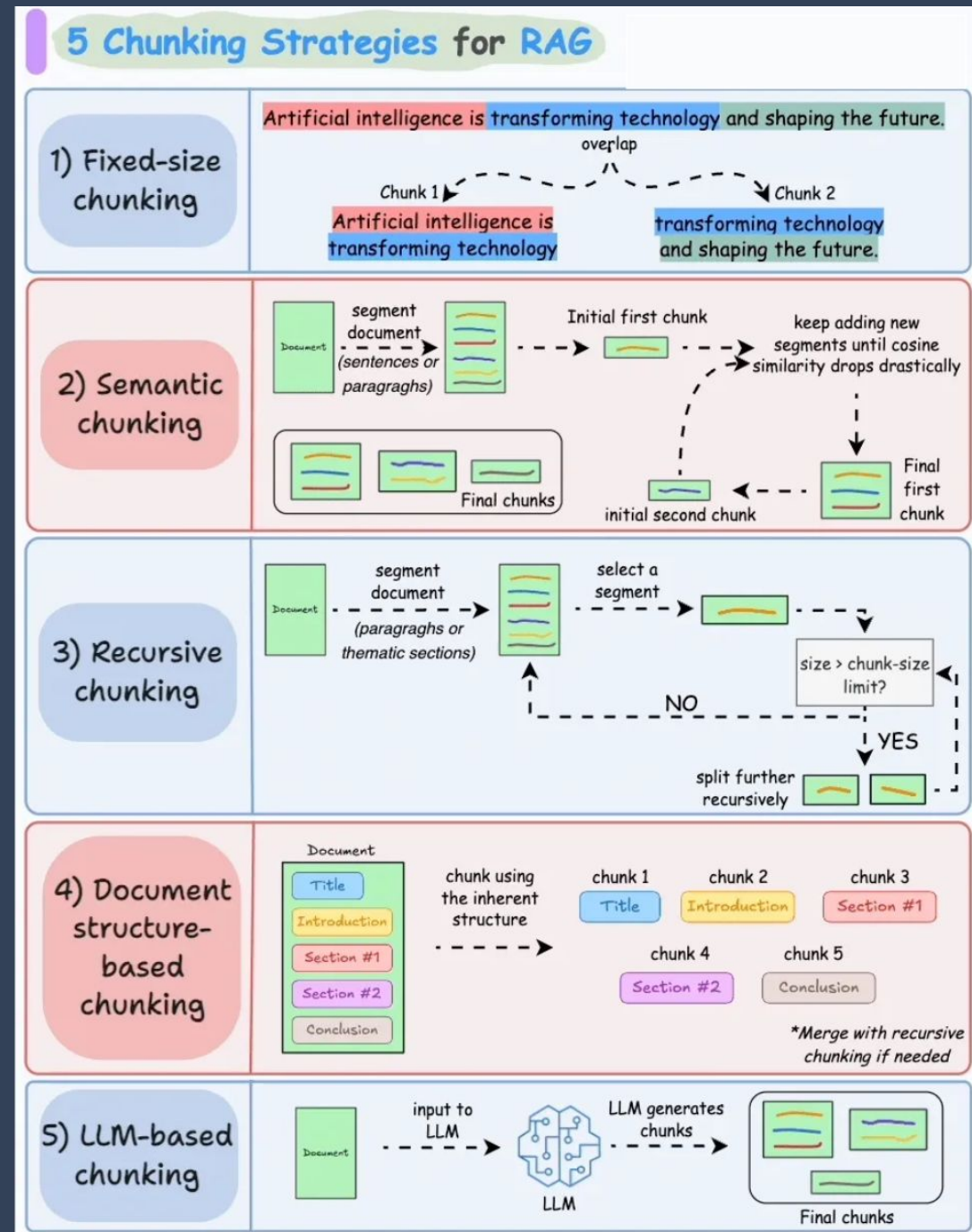
5) LLM-based chunking



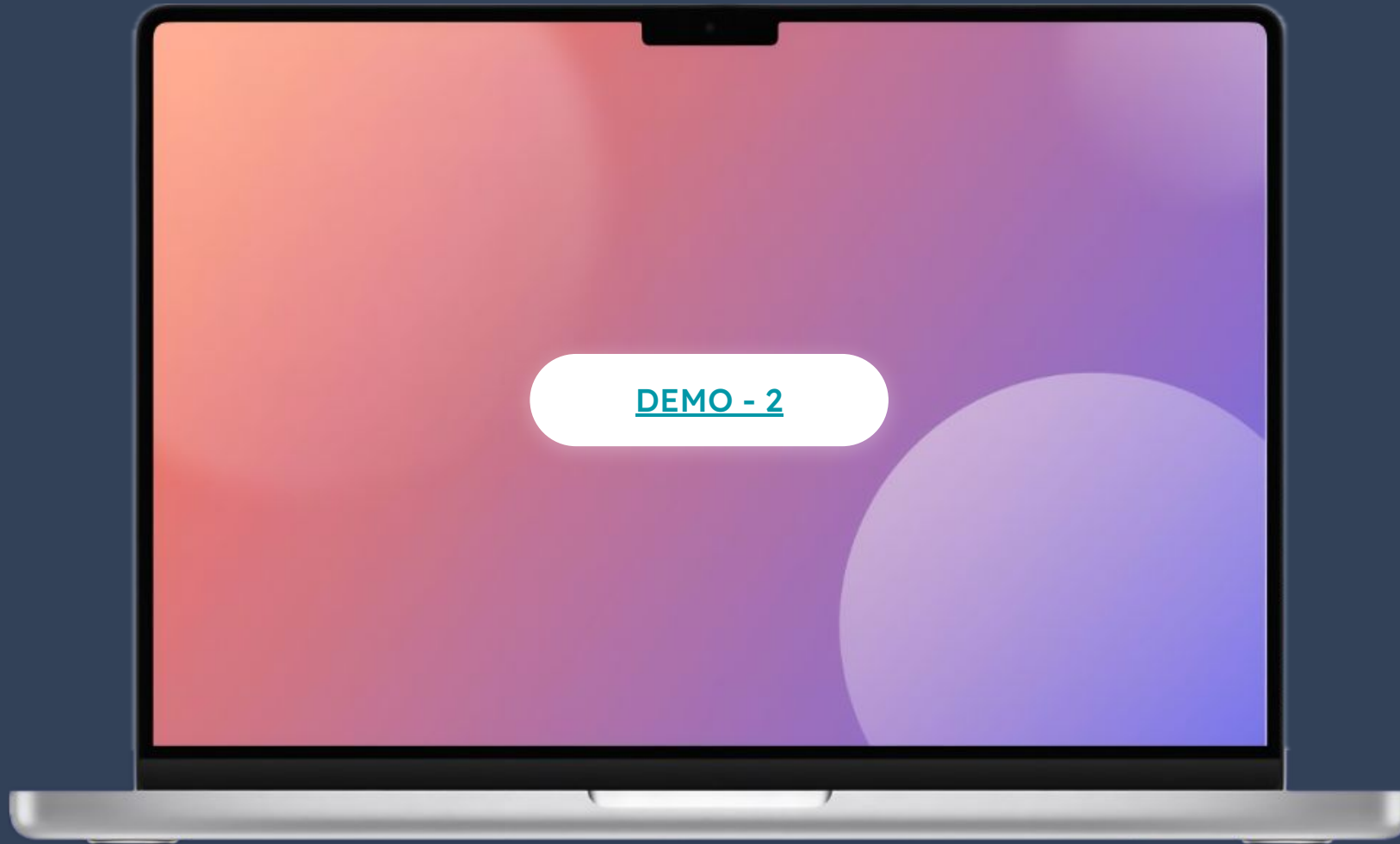
- Since every approach has upsides and downsides, why not use the LLM to create chunks?
- The LLM can be prompted to generate semantically isolated and meaningful chunks.
- Quite evidently, this method will ensure high semantic accuracy since the LLM can understand context and meaning beyond simple heuristics (used in the above four approaches).

Cheat Sheet : 5 chunking strategies for RAG

👉 Refer back to this slide during revision to choose the right chunking strategy



Module 2 - Implementation Notes: Chunking Strategies



No Question Is Too Small Every Question Deserves an Answer!!

Ask about anything:

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Quick Request

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Live Zoom Poll



We are launching a poll on Zoom. Please take 2 minutes to fill out the form.

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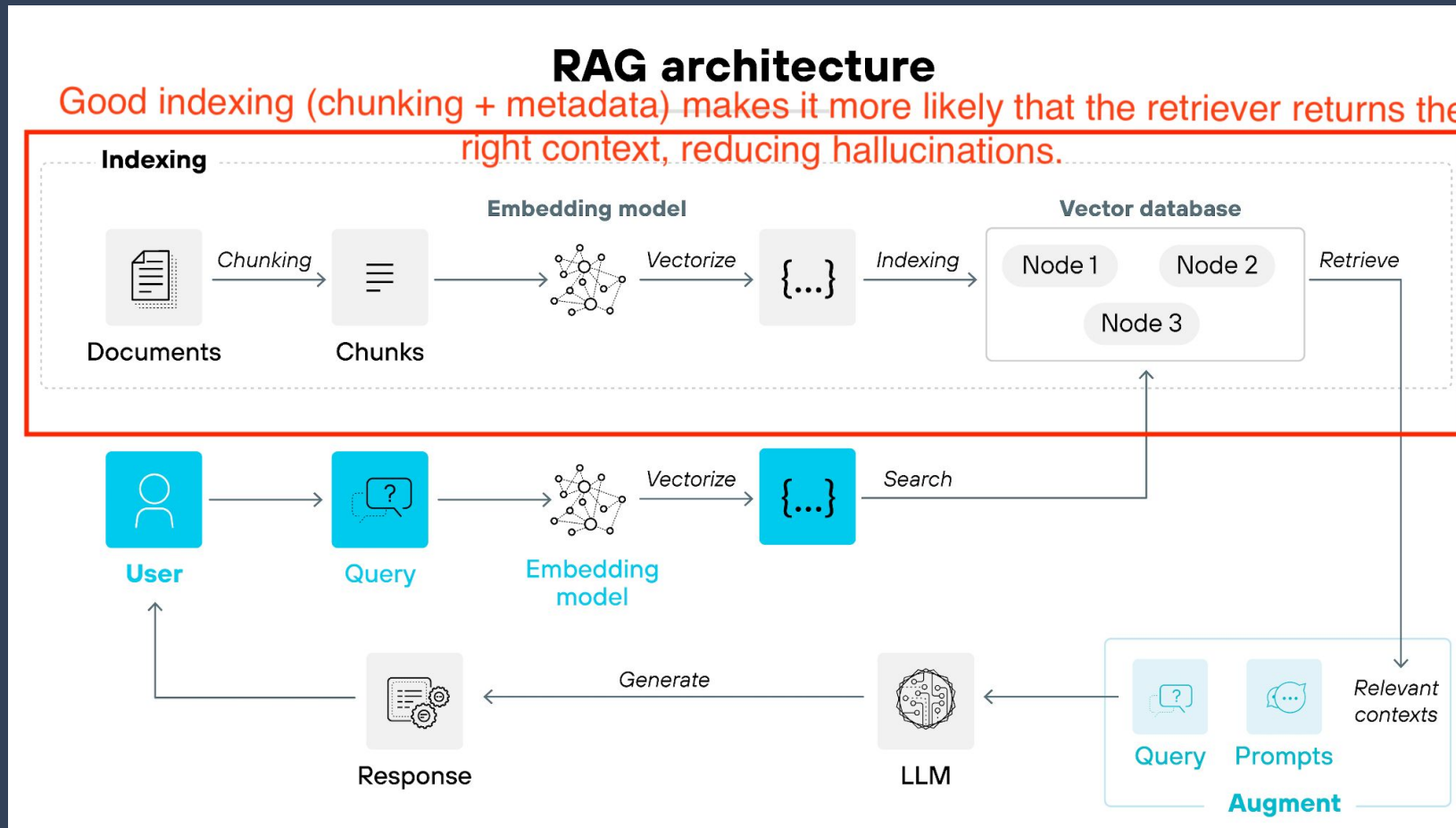


Indexing
Strategies
for RAG



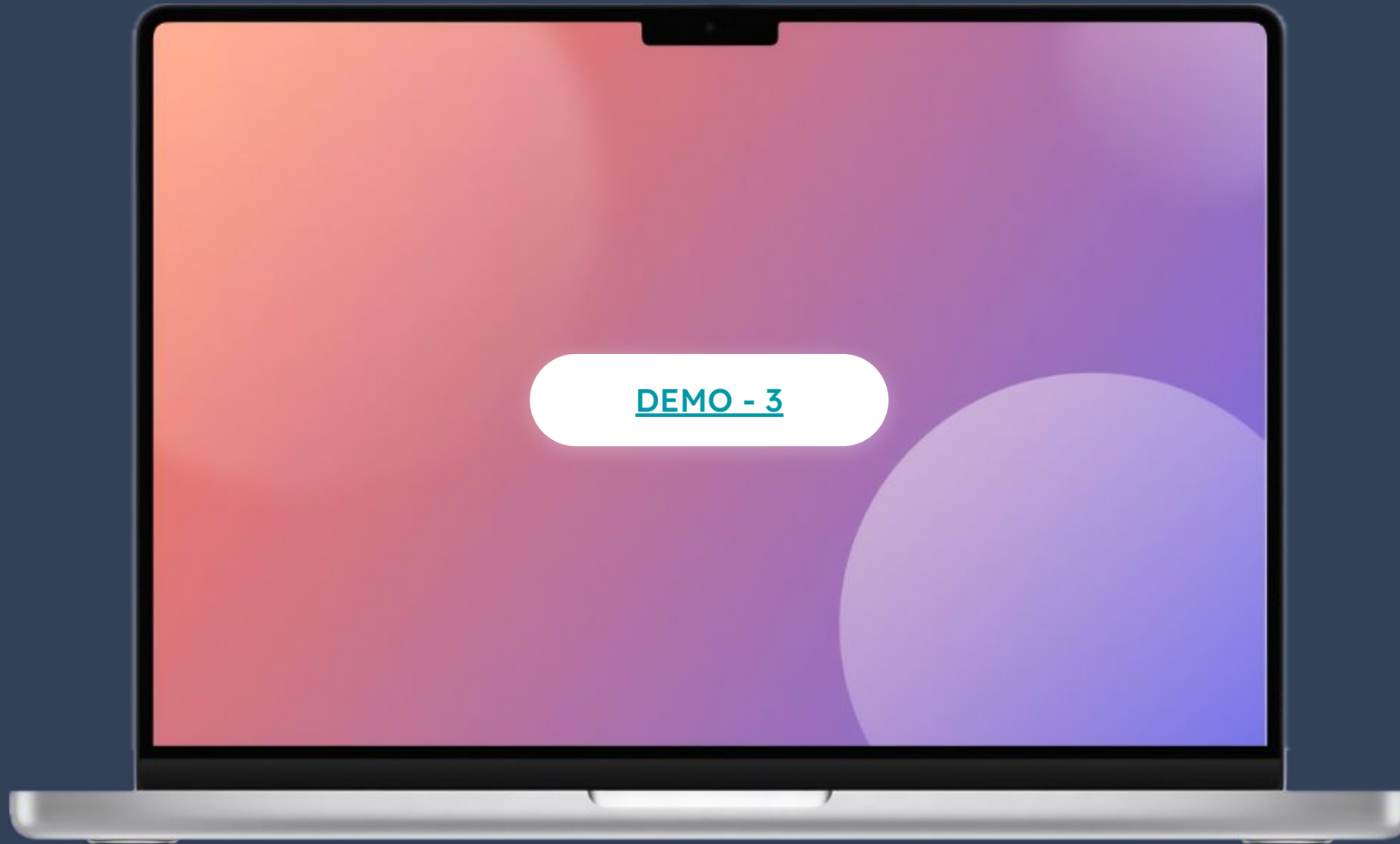
Why Indexing is Required in RAG

- **RAG needs a fast way to find relevant evidence.** Indexing converts documents into **retrievable units (chunks/nodes)** with embeddings + metadata.
- **Without indexing, retrieval is too slow at scale.** You'd be doing a near **full scan** over all chunks every query (high latency, high cost).



Indexing happens **offline**; retrieval happens **online at query time**

Module 3 - Implementation Notes: Indexing Strategies for RAG



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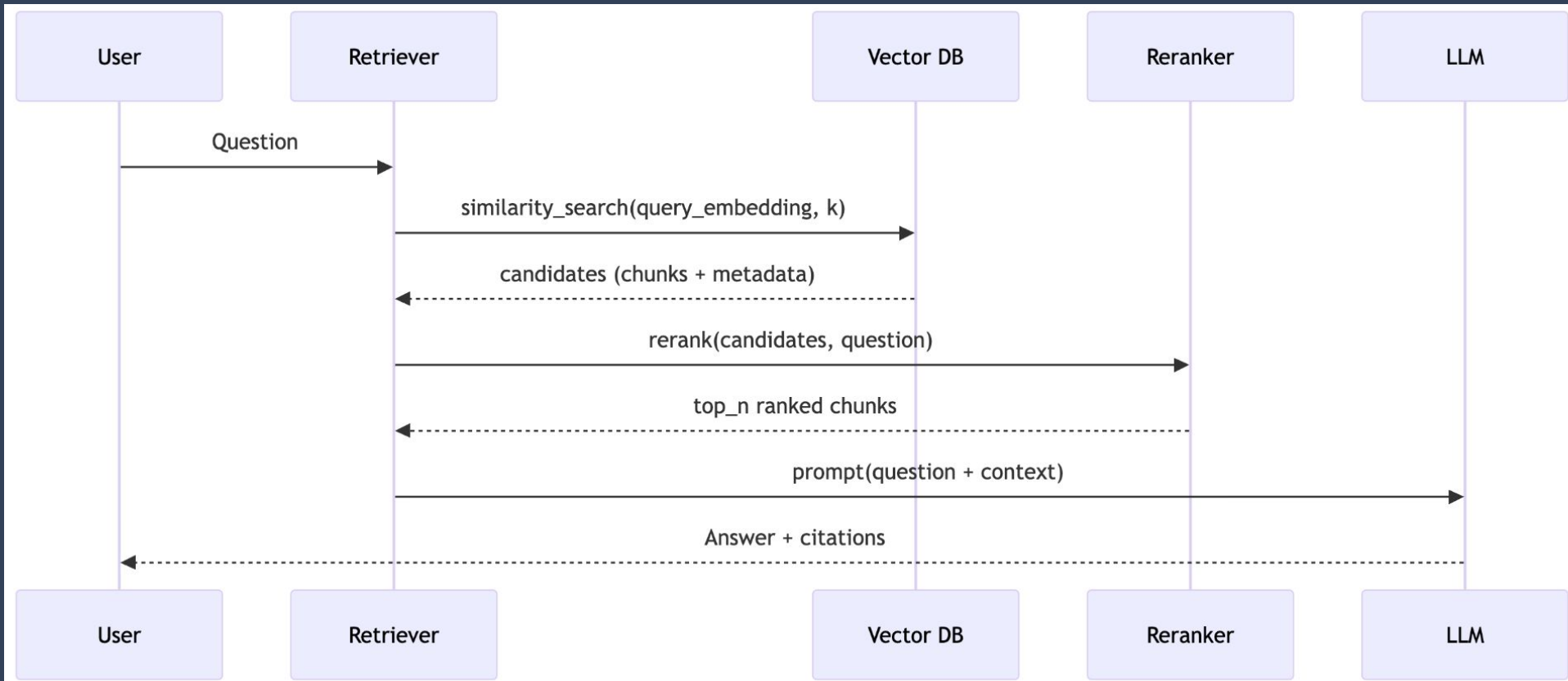
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Retrieval in One Picture (The Core Loop)

- Convert question → searchable representation
- Fetch candidates → Top-K chunks
- Improve relevance → rerank
- Send best context → grounded answer



Week 2 Summary: From Embeddings → Indexing

